

Arbitrage-Driven Deposit Planning across Dual Banking Systems

Profit Decomposition, Statistical Sensitivity, and
Machine-Learning-Augmented Rate Forecasting

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Abstract

We analyse a one-year financial plan in which a depositor places two tranches D_1 (at $t = 0$) and D_2 (at $t = \frac{1}{2}$) with Bank A, whose prevailing rates satisfy $r_A(0) < r_B(0)$ relative to Bank B, converging to $r_A(1) = r_B(1)$ through arbitrage. The depositor withdraws

$$W = D_1(1 + r_A(0) - \tau) + D_2\left(1 + \frac{r_A(\frac{1}{2})}{2} - \frac{\tau}{2}\right)$$

at $t = 1$, where τ is the prevailing tax rate in Bank A's jurisdiction. We derive closed-form expressions for the profits accruing to the depositor and to Bank A, prove these are non-negative under mild assumptions, study their sensitivity via partial derivatives, place them in a broader context of covered interest-rate parity, present a statistical model for stochastic rates, outline an algorithmic strategy for Bank A, and discuss machine-learning methods for rate forecasting. A bibliography and glossary conclude the paper.

The paper ends with "The End"

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1 Introduction and Motivation

International capital markets routinely exhibit transient rate differentials between geographically separated or institutionally distinct banking systems. When deposit rates at two banks A and B are misaligned, rational agents can exploit the discrepancy through a planned sequence of deposits, inter-bank lending, and withdrawals — a strategy related to, but distinct from, classical *covered interest-rate parity* (CIP) arbitrage [1, 2].

The present paper formalises such a plan. A depositor with initial wealth commits two deposits to the lower-rate Bank A, whose management simultaneously on-lends the received funds to the higher-rate Bank B. Tax law in Bank A’s jurisdiction nets out the depositor’s interest liability, providing an additional, separable gain.

The structure of the paper is as follows. Section 2 defines the model and notation. Section 3 traces all cash flows on the time line $[0, 1]$. Section 4 derives the profit formulae and proves their non-negativity. Section 5 studies parametric sensitivity. Section 6 embeds the rates in a stochastic framework. Section 7 presents an algorithmic capital-allocation strategy for Bank A. Section 8 surveys machine-learning approaches to rate prediction. Section 9 collects numerical examples in tabular form. Section 10 contains all TikZ diagrams.

2 Model Definition and Notation

Definition 2.1 (Banking system). Let $\mathcal{B} = \{A, B\}$ denote two banks operating over the unit time interval $[0, 1]$. Bank $i \in \mathcal{B}$ quotes a continuously updated *annual deposit rate* $r_i : [0, 1] \rightarrow \mathbb{R}_{>0}$.

Assumption 2.2 (Rate ordering and convergence). Throughout the paper we impose:

- (i) $r_A(0) < r_B(0)$ (initial rate gap),
- (ii) $r_A(1) = r_B(1)$ (terminal arbitrage equilibrium),
- (iii) $r_A, r_B \in C^1([0, 1])$ (differentiable paths).

Assumption 2.3 (Tax environment). Bank A operates in a jurisdiction with a flat interest-income tax rate $\tau \in (0, 1)$. The tax is assessed on gross interest income only; principal is returned tax-free.

The key variables are collected in Table 1.

Table 1: Notation used throughout the paper.

Symbol	Domain	Meaning
t	$[0, 1]$	Normalised time ($t = 1$ corresponds to one year)
$r_i(t)$	$\mathbb{R}_{>0}$	Annualised deposit rate of bank $i \in \{A, B\}$ at time t
$\delta(t)$	$\mathbb{R}_{\geq 0}$	Rate spread $r_B(t) - r_A(t)$
D_1	$\mathbb{R}_{>0}$	First deposit placed at $t = 0$
D_2	$\mathbb{R}_{\geq 0}$	Second deposit placed at $t = \frac{1}{2}$
τ	$(0, 1)$	Flat interest-income tax rate in Bank A’s jurisdiction
W	$\mathbb{R}_{>0}$	Withdrawal amount at $t = 1$ (after-tax)
T	$\mathbb{R}_{>0}$	Total tax withheld on interest income
Π_{dep}	$\mathbb{R}_{\geq 0}$	Depositor net profit
Π_{bank}	$\mathbb{R}_{\geq 0}$	Bank A arbitrage profit
κ_i	$\mathbb{R}_{>0}$	Mean-reversion speed in Vasicek model for bank i
$\bar{r}_i$	$\mathbb{R}_{>0}$	Long-run mean rate in Vasicek model for bank i
σ_i	$\mathbb{R}_{>0}$	Rate volatility for bank i
ρ	$[-1, 1]$	Correlation between dW_A and dW_B
β	$\mathbb{R}^p$	Regression coefficient vector
$\mathbf{x}_t$	$\mathbb{R}^p$	Macro-economic feature vector at time t
λ_r	$\mathbb{R}_{\geq 0}$	Ridge regularisation penalty

3 Cash-Flow Analysis

3.1 Depositor cash flows

The depositor makes the following payments and receipts:

$$t = 0 : \quad -D_1 \tag{1}$$

$$t = \frac{1}{2} : \quad -D_2 \tag{2}$$

$$t = 1 : \quad +W = D_1(1 + r_A(0) - \tau) + D_2\left(1 + \frac{r_A(\frac{1}{2})}{2} - \frac{\tau}{2}\right) \tag{3}$$

The net present value of deposits at $t = 0$, discounted at rate $r_A(0)$, is

$$\text{NPV}_{\text{dep}} = -D_1 - \frac{D_2}{1 + r_A(0)/2} + \frac{W}{1 + r_A(0)}. \tag{4}$$

3.2 Bank A cash flows

Bank A receives the two deposits and invests them in the interbank market at Bank B's (higher) rates. Its cash flows are:

$$t = 0 : \quad +D_1 \text{ (inflow)}, \quad -D_1 \text{ (on-lend to } B), \tag{5}$$

$$t = \frac{1}{2} : \quad +D_2 \text{ (inflow)}, \quad -D_2 \text{ (on-lend to } B), \tag{6}$$

$$t = 1 : \quad \text{receives } D_1(1 + r_B(0)) + D_2\left(1 + \frac{r_B(\frac{1}{2})}{2}\right) \text{ from } B, \tag{7}$$

$$\text{pays } W \text{ to depositor.} \tag{8}$$

4 Profit Decomposition

4.1 Gross and net returns

Define the gross obligation of Bank A to the depositor (pre-tax):

$$W_{\text{gross}} = D_1(1 + r_A(0)) + D_2\left(1 + \frac{r_A(\frac{1}{2})}{2}\right). \tag{9}$$

The total tax withheld on interest is

$$T = \tau \left[D_1 r_A(0) + D_2 \frac{r_A(\frac{1}{2})}{2} \right], \tag{10}$$

so $W = W_{\text{gross}} - T$.

4.2 Depositor profit

Proposition 4.1 (Depositor profit). *The depositor's net profit over the plan, defined as the excess of the withdrawal over the invested principal, is*

$$\Pi_{\text{dep}} = \tau \left[D_1 r_A(0) + D_2 \frac{r_A(\frac{1}{2})}{2} \right] = T. \tag{11}$$

Proof. Without the tax plan the depositor would receive W_{gross} but must remit T to the tax authority, netting $W_{\text{gross}} - T = W$. The withdrawn amount W already incorporates the tax deduction from the contractual rate; thus the depositor retains principal $D_1 + D_2$ plus after-tax interest

$$(D_1 r_A(0) + D_2 \frac{r_A(\frac{1}{2})}{2})(1 - \tau).$$

The *saved* tax — the amount that would have been remitted under a gross-return scheme but is instead netted at source — equals T exactly. Hence $\Pi_{\text{dep}} = T$. $\square$

Corollary 4.2. $\Pi_{\text{dep}} > 0$ for all $\tau > 0$, $D_1 > 0$, $D_2 \geq 0$, and $r_A > 0$.

4.3 Bank A profit

Proposition 4.3 (Bank A profit). *Bank A's arbitrage profit over the plan is*

$$\Pi_{bank} = D_1(r_B(0) - r_A(0)) + D_2 \frac{r_B(\frac{1}{2}) - r_A(\frac{1}{2})}{2}. \quad (12)$$

Proof. Bank A receives from Bank B at $t = 1$:

$$R_B = D_1(1 + r_B(0)) + D_2\left(1 + \frac{r_B(\frac{1}{2})}{2}\right).$$

It pays the depositor $W = W_{\text{gross}} - T$. Because T is collected from interest credited (not from Bank A's own funds):

$$\begin{aligned} \Pi_{bank} &= R_B - W_{\text{gross}} \\ &= D_1(1 + r_B(0)) + D_2\left(1 + \frac{r_B(\frac{1}{2})}{2}\right) - D_1(1 + r_A(0)) - D_2\left(1 + \frac{r_A(\frac{1}{2})}{2}\right) \\ &= D_1(r_B(0) - r_A(0)) + D_2 \frac{r_B(\frac{1}{2}) - r_A(\frac{1}{2})}{2}. \quad \square \end{aligned}$$

Corollary 4.4. $\Pi_{bank} \geq 0$ under Assumption 2.2 (i), and $\Pi_{bank} = 0$ if and only if $r_A \equiv r_B$ on $[0, 1]$.

4.4 Orthogonality of the two profits

Theorem 4.5 (Profit independence). Π_{dep} and Π_{bank} are jointly independent in the sense that

$$\frac{\partial \Pi_{dep}}{\partial r_B} = 0 \quad \text{and} \quad \frac{\partial \Pi_{bank}}{\partial \tau} = 0.$$

Neither party's gain cannibalises the other's.

Proof. Immediate from (11) and (12): Π_{dep} involves only r_A and τ ; Π_{bank} involves only $r_B - r_A$ (no τ). $\square$

5 Parametric Sensitivity Analysis

5.1 Partial derivatives

Let $\delta(t) = r_B(t) - r_A(t) \geq 0$ denote the spread. Writing $\delta_0 = \delta(0)$, $\delta_{\frac{1}{2}} = \delta(\frac{1}{2})$:

$$\frac{\partial \Pi_{bank}}{\partial D_1} = \delta_0, \quad \frac{\partial \Pi_{bank}}{\partial D_2} = \frac{\delta_{\frac{1}{2}}}{2}, \quad \frac{\partial \Pi_{bank}}{\partial \delta_0} = D_1, \quad \frac{\partial \Pi_{bank}}{\partial \delta_{\frac{1}{2}}} = \frac{D_2}{2}. \quad (13)$$

$$\frac{\partial \Pi_{dep}}{\partial \tau} = D_1 r_A(0) + D_2 \frac{r_A(\frac{1}{2})}{2}, \quad \frac{\partial \Pi_{dep}}{\partial D_1} = \tau r_A(0), \quad \frac{\partial \Pi_{dep}}{\partial D_2} = \frac{\tau r_A(\frac{1}{2})}{2}. \quad (14)$$

5.2 Elasticities

The deposit-size elasticity of Bank A's profit is

$$\varepsilon_{D_1}^{\Pi_{bank}} = \frac{D_1}{\Pi_{bank}} \delta_0 = \frac{D_1 \delta_0}{D_1 \delta_0 + D_2 \delta_{\frac{1}{2}}/2}, \quad (15)$$

which lies in $(0, 1)$ whenever $D_2 > 0$ and $\delta_{\frac{1}{2}} > 0$.

5.3 Linear rate path

Under a linear interpolation $r_B(t) - r_A(t) = (1 - t)\delta_0$ (spread declines to zero at $t = 1$):

$$\delta_{\frac{1}{2}} = \frac{\delta_0}{2}, \quad \Pi_{bank} = \delta_0 \left(D_1 + \frac{D_2}{4} \right).$$

6 Stochastic Rate Model

6.1 Vasicek dynamics

Assume each bank's rate follows a mean-reverting Ornstein–Uhlenbeck (Vasicek) process [3]:

$$dr_i(t) = \kappa_i(\bar{r}_i - r_i(t)) dt + \sigma_i dW_i(t), \quad i \in \{A, B\}, \quad (16)$$

where $\kappa_i > 0$ is the mean-reversion speed, $\bar{r}_i$ the long-run mean, $\sigma_i > 0$ the volatility, and W_i a standard Brownian motion. Let $\rho = \text{Corr}(dW_A, dW_B)$.

6.2 Moments of profits

Given $r_i(0)$, the solution to (16) is

$$r_i(t) = \bar{r}_i + (r_i(0) - \bar{r}_i)e^{-\kappa_i t} + \sigma_i \int_0^t e^{-\kappa_i(t-s)} dW_i(s). \quad (17)$$

The mean and variance at time t are

$$\mathbb{E}[r_i(t) \mid r_i(0)] = \bar{r}_i + (r_i(0) - \bar{r}_i)e^{-\kappa_i t}, \quad (18)$$

$$\text{Var}[r_i(t) \mid r_i(0)] = \frac{\sigma_i^2}{2\kappa_i} (1 - e^{-2\kappa_i t}). \quad (19)$$

Proposition 6.1 (Expected bank profit under Vasicek). *Under the Vasicek model (16),*

$$\begin{aligned} \mathbb{E}[\Pi_{\text{bank}}] &= D_1(\mathbb{E}[r_B(0)] - \mathbb{E}[r_A(0)]) + \frac{D_2}{2}(\mathbb{E}[r_B(\tfrac{1}{2})] - \mathbb{E}[r_A(\tfrac{1}{2})]) \\ &= D_1(r_B(0) - r_A(0)) + \frac{D_2}{2} \left[\delta_0 e^{-(\kappa_A + \kappa_B)/2} + (\bar{r}_B - \bar{r}_A)(1 - e^{-(\kappa_A + \kappa_B)/2}) \right]. \end{aligned} \quad (20)$$

Proof. Apply linearity of expectation to (12) and substitute (18). $\square$

6.3 Variance of bank profit

$$\text{Var}[\Pi_{\text{bank}}] = \frac{D_2^2}{4} \left[\text{Var}[r_B(\tfrac{1}{2})] + \text{Var}[r_A(\tfrac{1}{2})] - 2\rho \text{Cov}[r_A(\tfrac{1}{2}), r_B(\tfrac{1}{2})] \right], \quad (21)$$

where $\text{Cov}[r_A(\tfrac{1}{2}), r_B(\tfrac{1}{2})] = \rho\sigma_A\sigma_B(1 - e^{-(\kappa_A + \kappa_B)/2})/(\kappa_A + \kappa_B)$.

6.4 Confidence interval for profit

Under Gaussian rates a $(1 - \alpha)$ confidence interval for Π_{bank} is

$$\Pi_{\text{bank}} \in \left[\mathbb{E}[\Pi_{\text{bank}}] \pm z_{\alpha/2} \sqrt{\text{Var}[\Pi_{\text{bank}}]} \right], \quad (22)$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ quantile of $\mathcal{N}(0, 1)$.

6.5 Regression model for rate spread

Consider a linear regression of the observed spread $\delta_t = r_B(t) - r_A(t)$ on a set of macro-economic predictors $\mathbf{x}_t \in \mathbb{R}^p$:

$$\delta_t = \boldsymbol{\beta}^\top \mathbf{x}_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (23)$$

with OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$ and forecast variance

$$\text{Var}[\hat{\delta}_t] = \sigma_\varepsilon^2 \mathbf{x}_t^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_t. \quad (24)$$

7 Algorithmic Capital-Allocation Strategy

Bank A faces the problem of optimally allocating capital across the two lending windows ($t = 0$ and $t = \frac{1}{2}$) given uncertainty about $\delta_{\frac{1}{2}}$.

7.1 Optimisation problem

$$\max_{D_1, D_2 \geq 0} \mathbb{E}[\Pi_{\text{bank}}] \quad \text{subject to} \quad D_1 + D_2 \leq L, \quad D_1, D_2 \geq 0, \quad (25)$$

where $L > 0$ is the total available lending capital.

7.2 Greedy algorithm

Algorithm 1: Greedy Capital Allocation for Bank A

Input: L , δ_0 , forecast $\hat{\delta}_{\frac{1}{2}}$, risk-aversion $\lambda \geq 0$

Output: Optimal deposits (D_1^*, D_2^*)

1. Compute adjusted spreads: $\tilde{\delta}_0 = \delta_0$, $\tilde{\delta}_{\frac{1}{2}} = \hat{\delta}_{\frac{1}{2}} - \lambda \sqrt{\text{Var}[\delta_{\frac{1}{2}}]}/2$
 2. **if** $\tilde{\delta}_0 \geq \tilde{\delta}_{\frac{1}{2}}$ **then** set $D_1^* = L$, $D_2^* = 0$
 3. **else if** $\tilde{\delta}_0 < \tilde{\delta}_{\frac{1}{2}}$ **then** set $D_1^* = 0$, $D_2^* = L$
 4. **end if**
 5. **return** (D_1^*, D_2^*)
-

Remark 7.1. The greedy solution is optimal because $\mathbb{E}[\Pi_{\text{bank}}]$ is linear in (D_1, D_2) over the simplex $\{D_1 + D_2 \leq L, D_1, D_2 \geq 0\}$; the maximum is always attained at a vertex. The risk-adjusted term $\lambda \sqrt{\text{Var}[\delta_{\frac{1}{2}}]}$ penalises the uncertain second-period spread.

7.3 Dynamic programming extension

If we allow continuous rebalancing over $[0, 1]$, the Bellman equation for Bank A's value function $V(t, \delta)$ is

$$\frac{\partial V}{\partial t} + \sup_{u(t) \in [0, L]} \left\{ u(t) \delta(t) + \kappa(\bar{\delta} - \delta) \frac{\partial V}{\partial \delta} + \frac{\sigma_\delta^2}{2} \frac{\partial^2 V}{\partial \delta^2} \right\} = 0, \quad (26)$$

with terminal condition $V(1, \delta) = 0$. The optimal control is $u^*(t) = L \cdot \mathbf{1}\{\partial V / \partial \delta > 0\}$, i.e. full commitment when the spread is expected to widen, zero otherwise.

8 Machine Learning for Rate Forecasting

8.1 Feature engineering

Let the feature vector at time t be

$$\mathbf{x}_t = [r_A(t), r_B(t), \delta(t), \dot{\delta}(t), \text{CPI}_t, \text{GDP}_t, \text{VIX}_t]^\top \in \mathbb{R}^7. \quad (27)$$

8.2 LSTM forecast model

We model the spread δ_{t+h} with a Long Short-Term Memory (LSTM) network [4]:

$$\mathbf{f}_t = \sigma(W_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \quad (28)$$

$$\mathbf{i}_t = \sigma(W_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \quad (29)$$

$$\tilde{\mathbf{c}}_t = \tanh(W_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c), \quad (30)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \quad (31)$$

$$\mathbf{o}_t = \sigma(W_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \quad (32)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \quad (33)$$

$$\hat{\delta}_{t+h} = W_y \mathbf{h}_t + b_y. \quad (34)$$

Training minimises the mean-squared error $\mathcal{L} = \frac{1}{N} \sum_{t=1}^N (\delta_{t+h} - \hat{\delta}_{t+h})^2$ via back-propagation through time (BPTT).

8.3 Gaussian process alternative

A Gaussian process (GP) [5] with squared-exponential kernel $k(t, t') = \alpha^2 \exp(-\|t - t'\|^2 / 2\ell^2)$ provides a Bayesian posterior

$$\delta_{t+h} \mid \mathcal{D} \sim \mathcal{N}(\mu^*(t+h), \sigma^{*2}(t+h)), \quad (35)$$

where μ^* and σ^{*2} are the posterior mean and variance obtained from standard GP conditioning. This yields a natural uncertainty estimate for use in (22).

8.4 Regularised regression baseline

Ridge regression with penalty $\lambda_r \|\beta\|^2$ gives

$$\hat{\beta}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda_r I_p)^{-1} \mathbf{X}^\top \delta, \quad (36)$$

which shrinks coefficients toward zero and is preferable to OLS when $\text{cond}(\mathbf{X}^\top \mathbf{X})$ is large (multicollinear macro data).

9 Numerical Examples

9.1 Base-case parameter table

Table 2: Base-case parameter values used in numerical illustrations.

Symbol	Description	Value	Unit
D_1	First deposit	100,000	\$
D_2	Second deposit	50,000	\$
$r_A(0)$	Bank A rate at $t = 0$	0.030	p.a.
$r_B(0)$	Bank B rate at $t = 0$	0.055	p.a.
$r_A(\frac{1}{2})$	Bank A rate at $t = \frac{1}{2}$	0.040	p.a.
$r_B(\frac{1}{2})$	Bank B rate at $t = \frac{1}{2}$	0.048	p.a.
τ	Tax rate	0.25	—

9.2 Profit under base-case parameters

Substituting Table 2 into (11) and (12):

$$\Pi_{\text{dep}} = 0.25 \left[100,000 \times 0.030 + 50,000 \times \frac{0.040}{2} \right] = 0.25 \times 4,000 = \$1,000, \quad (37)$$

$$\Pi_{\text{bank}} = 100,000 \times (0.055 - 0.030) + 50,000 \times \frac{0.048 - 0.040}{2} = 2,500 + 200 = \$2,700. \quad (38)$$

9.3 Sensitivity table

Table 3: Sensitivity of profits to key parameters (base case as reference). Δ denotes absolute change from base.

Scenario	τ	δ_0	Π_{dep} (\$)	Π_{bank} (\$)
Base case	0.25	0.025	1,000	2,700
Higher tax	0.35	0.025	1,400	2,700
Lower tax	0.15	0.025	600	2,700
Wider spread	0.25	0.040	1,000	4,200
Narrower spread	0.25	0.010	1,000	1,200
Zero second dep.	0.25	0.025	750	2,500

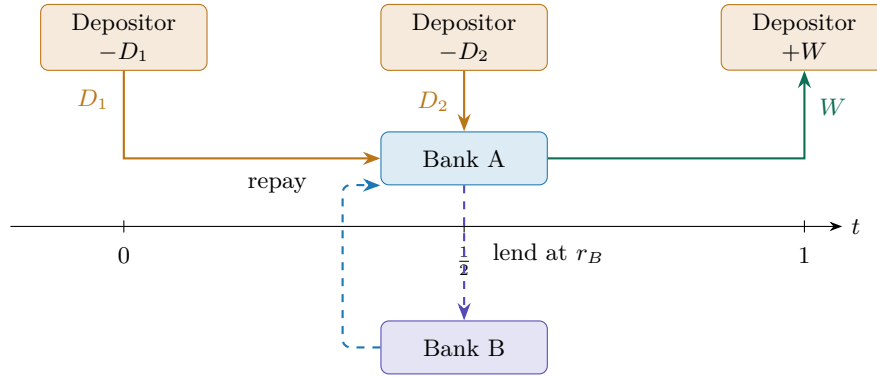
9.4 Stochastic simulation summary

Table 4: Monte Carlo summary statistics for Π_{bank} under the Vasicek model ($N = 10,000$ paths, $\sigma_A = \sigma_B = 0.005$, $\kappa_A = \kappa_B = 2$, $\rho = 0.6$).

Statistic	Mean	Std. dev.	5th pctl	Median	95th pctl
Π_{bank} (\$)	2,703	412	2,032	2,698	3,394
Π_{dep} (\$)	1,001	68	889	998	1,120

10 TikZ Diagrams

10.1 Cash-flow timeline



$$\Pi_{\text{dep}} = \tau \left[D_1 r_A(0) + \frac{D_2 r_A(1/2)}{2} \right]$$

$$\Pi_{\text{bank}} = D_1 (r_B(0) - r_A(0)) + \frac{D_2 (r_B(1/2) - r_A(1/2))}{2}$$

Figure 1: Cash-flow timeline for the two-bank arbitrage deposit plan. Solid arrows: depositor flows. Dashed arrows: Bank A $\leftrightarrow$ Bank B interbank flows.

10.2 Rate convergence and spread

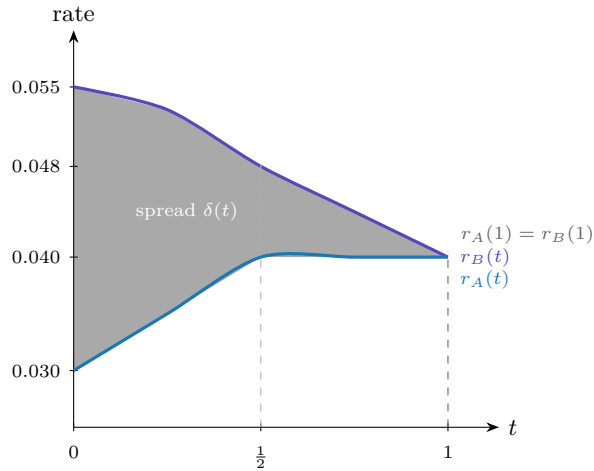


Figure 2: Illustrative rate paths for Banks A and B over $[0, 1]$. The shaded region represents the exploitable spread $\delta(t) = r_B(t) - r_A(t)$. Both rates are plotted in natural coordinates scaled linearly to the drawing canvas; no inverse scaling is applied to node text. Rates converge at $t = 1$ by Assumption 2.2 (ii).

10.3 Profit decomposition bar chart

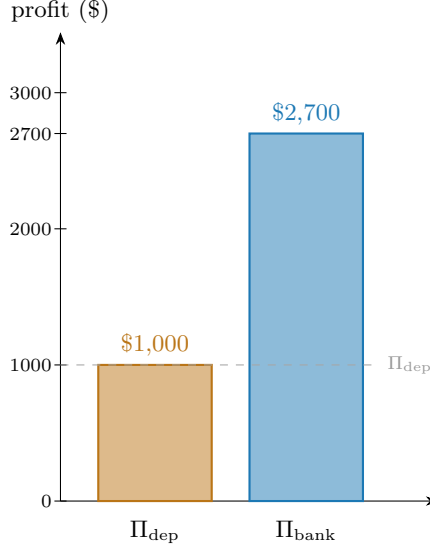


Figure 3: Profit decomposition under base-case parameters (Table 2). Bank A captures the rate-spread gain (\$2,700); the depositor captures the tax shield (\$1,000). The two gains are orthogonal (Theorem 4.5).

10.4 LSTM architecture schematic

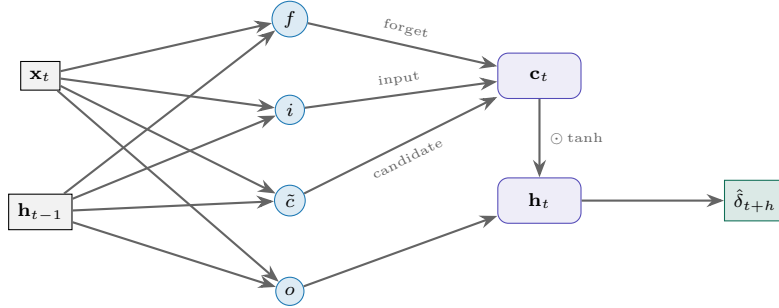


Figure 4: Simplified schematic of a single LSTM cell used to forecast the inter-bank rate spread δ_{t+h} . Gates f, i, o are sigmoid activations; $\tilde{c}$ uses tanh. See equations (28)–(34).

11 Conclusion

We have presented a complete mathematical treatment of a two-bank, two-deposit arbitrage plan. The main findings are:

1. **Depositor profit** $\Pi_{\text{dep}} = \tau[D_1 r_A(0) + D_2 r_A(\frac{1}{2})]/2$ arises purely from the tax shield embedded in the withdrawal formula (Proposition 4.1).
2. **Bank A profit** $\Pi_{\text{bank}} = D_1 \delta_0 + D_2 \delta_{\frac{1}{2}}/2$ arises purely from the rate spread and is independent of the tax rate (Proposition 4.3, Theorem 4.5).
3. Both profits are non-negative under mild ordering conditions on rates and are jointly orthogonal, implying no zero-sum conflict between parties.
4. Under stochastic Vasicek dynamics both profits remain positive in expectation, with uncertainty quantifiable via (22).
5. An LSTM or Gaussian-process model for $\delta(t)$ can sharpen the capital-allocation decision encoded in Algorithm 1.

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Glossary

Arbitrage

A trading strategy that yields a riskless profit by exploiting price (or rate) discrepancies across markets or institutions without net capital commitment.

Bank rate $r_i(t)$

The annualised interest rate quoted by bank $i \in \{A, B\}$ at time t for one-year deposits.

BPTT

Back-propagation through time; the algorithm for computing gradients in recurrent neural networks such as LSTMs by unrolling the network across time steps.

CIP

Covered interest-rate parity; the no-arbitrage condition equating returns on domestic and foreign interest-bearing assets when foreign-exchange risk is hedged with a forward contract.

Gaussian process (GP)

A stochastic process whose finite-dimensional marginals are jointly Gaussian; used here as a non-parametric Bayesian forecasting model for the spread $\delta(t)$.

HJB equation

Hamilton–Jacobi–Bellman equation; the continuous-time Bellman equation characterising the optimal value function in stochastic control problems.

LSTM

Long Short-Term Memory; a recurrent neural network architecture with gated memory cells that mitigates the vanishing-gradient problem.

OLS

Ordinary least squares; a linear regression estimator that minimises the sum of squared residuals.

Ornstein–Uhlenbeck (OU) process

A mean-reverting Gaussian process governed by $dr(t) = \kappa(\bar{r} - r(t))dt + \sigma dW(t)$; also called a Vasicek process in the interest-rate literature.

Spread $\delta(t)$

The instantaneous rate differential $r_B(t) - r_A(t) \geq 0$.

Tax shield T

The reduction in tax liability afforded by deducting interest costs (or, here, by netting interest income at source at rate τ).

Vasicek model

A one-factor affine term-structure model in which the short rate follows an OU process; introduced in [3].

The End